

Consilium

System Analysis Document (SAD)

AI Decision Copilot for Veterinary Clinics

UMHackathon 2026

1. Introduction

This application, Consilium, is a strategic Decision Intelligence Copilot engineered for small veterinary clinics (2–5 doctors). While standard practice management software focuses on administrative automation, Consilium operates at the cognitive layer of the clinic. It addresses the crisis of veterinary burnout and revenue leakage not by merely automating tasks, but by augmenting clinical decision-making at every stage of the patient journey.

Consilium provides decision support across three critical phases:

- **Contextual Intelligence:** synthesising years of unstructured history into an actionable brief.
- **Clinical Synthesis:** translating raw consultation dialogue (voice or text) and attached photos into structured medical logic, prescription, and billing intelligence.
- **Autonomous Triage:** using a high-fidelity reasoning engine with a tool-use loop to distinguish between routine recoveries and life-threatening post-surgical complications, ensuring that the right cases reach the doctor at the right time.

1.1 Purpose

This document outlines the technical architecture required to transition a clinic from manual oversight to an AI-assisted Decision Environment. It details how the system uses Anthropic Claude (Haiku 4.5 and Sonnet 4.6), a tool registry (Tavily web search and five user-facing clarifying tools), and a structured-output emit-tool pattern to ensure that every clinical decision — from billing to emergency escalation — is data-driven and consistent.

Key elements covered:

- **Architecture:** Consilium is a cloud-native platform built on a modular agentic framework. The system decouples the user interfaces (Next.js 16 dashboard and Telegram bot) from the core reasoning engine (Anthropic Claude orchestrated by a TypeScript tool-use loop in lib/llm.ts). A LangGraph Python sidecar is scaffolded for finals-round multi-turn checkpointing but is not on the critical path for the preliminary demo.
- **Data Flows:** the system transforms unstructured clinical signals (free text, dictated audio, attached photos, Telegram replies) into structured decision packets.
 - **Ingestion:** raw inputs (text, audio, images, or Telegram messages) are captured.
 - **Processing:** audio routes through Deepgram nova-3 for transcription; images upload to Supabase Storage; the combined payload flows through the API to

Claude, where it is cross-referenced with historical records and, if needed, fresh information from Tavily.

- **Persistence:** the resulting structured data (SOAP notes, billing flags, triage levels, conversation turns) is stored in PostgreSQL via Supabase.
- **Delivery:** critical updates are pushed to the doctor's dashboard via Supabase Realtime triggers to ensure immediate awareness.
- **Model Process:** the system maps the end-to-end Patient Decision Journey:
 - **Intelligence Briefing:** synthesising past visits into an actionable pre-consult summary via the emit_brief tool (Haiku 4.5).
 - **Clinical Extraction:** translating live consultation dialogue and photos into medical records and identifying revenue leakage via the emit_consult tool (Sonnet 4.6, with optional Tavily drug-recall lookups).
 - **Autonomous Surveillance:** monitoring pet recovery via Telegram post-discharge and using a triage tool-use loop with five clarifying tools and a final emit_decision tool (Sonnet 4.6) to escalate medical red flags to the vet.
- **Role of Reference:** this document serves as the technical blueprint for developers and a logic guide for stakeholders. It defines the API contracts, the database schema, and the agentic workflows required to ensure Claude provides consistent, high-accuracy clinical support across the clinic's operations.

1.2 Background

Small-to-medium veterinary clinics (2–5 doctors) in South-East Asia are currently bottlenecked by high-volume administrative overhead. Research shows vets lose over 3 hours daily to manual documentation, and clinics experience a 10–15% leakage in billable items due to fast-paced, unstructured consultations. Furthermore, once a pet is discharged, there is a clinical dark period where post-treatment complications go undetected until they become emergencies.

Previous version and current state:

Most clinics currently rely on fragmented Practice Management Systems (PMS) or legacy paper records. These systems act as static filing cabinets rather than active assistants. Communication is localised to the exam room, and follow-ups are either manual (phone calls) or non-existent, leading to poor owner compliance, missed secondary revenue, and no early warning for deteriorating patients.

Changes in major architectural components:

These five pillars move the system from a digital filing cabinet to an active assistant.

- **Reasoning Service Layer (Anthropic Claude):** the processor. Claude Sonnet 4.6 takes raw text plus optional photos and extracts professional medical notes (SOAP), a structured prescription, a billing recovery checklist, and a staff to-do list — all via a single `emit_consult` tool call. Haiku 4.5 powers the lighter-weight `emit_brief` path.
- **Tool-Use Triage Loop:** the brain. It remembers why the pet visited (e.g. surgery vs cough), may ask the owner exactly one targeted clarifying question (`request_photo`, `request_temperature`, `request_appetite_timeline`, `request_medication_compliance`, `schedule_doctor_callback`), and may consult Tavily once for drug-recall verification before committing a decision via `emit_decision`.
- **Telegram Integration:** the interface. Uses a common messaging app to automate follow-ups. No new app downloads required for owners. The `grammY` bot also forwards owner-sent photos through Claude vision.
- **Voice Capture (Deepgram nova-3):** the consult-room scribe. Browser MediaRecorder posts audio to `/api/transcribe`, returning a transcript with confidence in seconds. Auto-detects English, Bahasa Malaysia, and Mandarin code-switching common in SEA clinics.
- **Real-time Dashboard:** the hub. Uses Supabase Realtime to pop urgent alerts onto the doctor's screen the second Claude commits an escalation.

New capabilities:

- **Pre-Consult Briefs:** summarises years of messy medical history into 5 bullet points before the vet enters the room.
- **Multimodal Consult Capture:** voice dictation, typed notes, and attached wound/lab/X-ray photos all flow into one Claude call and emerge as structured SOAP, prescription, billing, and todos.
- **Billing Recovery Engine:** scans clinical notes for mentioned services (such as X-ray or e-collar) and flags them if they were not added to the invoice.
- **Automated Follow-Up Triage:** reaches out to owners via Telegram 24–48 hours post-visit and filters responses so the vet only sees the red flags.
- **Feedback Learning Loop:** captures doctor corrections and injects them as clinic-specific few-shot examples in future Claude calls, improving accuracy over time without model retraining.
- **Pet Health Passport:** an auto-generated, QR-accessible health record updated after every visit.

1.3 Target Stakeholders

Stakeholders	Roles	Expectations
Vet Doctor	Primary user. Receives pre-consult briefs, reviews and approves AI-generated clinical outputs, acknowledges escalation cards.	Reduce admin load by 3 hours/day. Catch 2–4 post-treatment complications early per month.
Pet Owner	End recipient. Interacts exclusively via Telegram bot — no dashboard access required.	>70% response rate on automated follow-ups. Instant health updates via QR passport link.
Clinic Manager	Monitors business health and revenue recovery via the monthly KPI panel on the dashboard.	Recover RM 8,000–12,000 in missed billing items monthly.
Development Team	Builds the platform, agentic logic, and database schema.	Modular codebase, reliable tool-use state across multi-turn Telegram conversations, clear API contracts, structured-output guardrails via emit-tool pattern.
QA Team	Validates system behaviour and AI accuracy.	>90% triage accuracy vs. keyword baseline. Full coverage of clinical edge cases including multimodal (photo + text) inputs and adversarial prompts.

2. System Architecture & Design

2.1 High-Level Architecture

2.1.1 Overview

Type	Details
System	Web Dashboard (Next.js 16) + Telegram Bot (grammY)
Architecture	Cloud-native monorepo with a decoupled reasoning engine
Deployment	Vercel (Next.js + API Routes), Supabase (PostgreSQL + Realtime + Storage)
AI Layer	Anthropic Claude — Haiku 4.5 (brief) and Sonnet 4.6 (consult, triage, multimodal vision). Mandatory for all reasoning, extraction, and triage decisions.
Speech-to-Text	Deepgram nova-3 — multilingual transcription for the consult room
Web Search	Tavily — server-side drug-recall and fresh-guidance lookups, 7-day cached
Tool Registry	lib/tools/registry.ts — per-feature tool exposure (server, user-facing, emit_*)
Agent Framework	TypeScript tool-use loop in lib/llm.ts (production). LangGraph (Python) with Supabase PostgresSaver scaffolded for finals-round multi-turn checkpointing.

Consilium is a client-server-agent system designed to act as a Decision Intelligence layer for the clinic. Rather than simple automation, the architecture provides draft decisions for the vet to validate. The system is composed of three primary intelligence components:

- **Triage Tool-Use Loop (lib/llm.ts + lib/telegram-handler.ts):** powers the Telegram bot (F3). It is a state-aware multi-turn loop that receives owner messages, may invoke exactly one clarifying tool when ambiguous, may consult Tavily for drug recalls, and emits a final decision (ALL_CLEAR, MONITOR, or ESCALATE) via emit_decision based on the specific procedure history.
- **Reasoning Layer (Anthropic Claude):** powers the clinical scribe (F1 and F2) and the triage decision (F3). Haiku 4.5 performs the speed-read brief synthesis; Sonnet 4.6

handles consult extraction with vision (wound, lab, X-ray photos) and triage classification with tool-use.

- **Persistence Layer (Supabase):** stores clinical data, conversation turns, and the corrections ledger, which captures doctor overrides to refine the prompt context for future reasoning tasks. Also hosts the tavily_cache (7-day TTL) and Storage buckets for consult-photos and owner-photos.

2.1.2 Claude as a Service Layer (The Decision Engine)

Claude is integrated as a dedicated service layer, called exclusively from server-side Next.js API routes. The Anthropic API key is never exposed to the client. Each feature constructs its own prompt chain and exposes its own tool subset:

- **Feature 1 (Brief, Haiku 4.5):** historical raw notes are chunked to the last 10 visits and injected into the BRIEF_PROMPT system context. Tools exposed: [emit_brief]. One forced tool call, one round trip, no web search.
- **Feature 2 (Consult, Sonnet 4.6):** raw consultation notes (typed or transcribed via Deepgram), patient metadata, the F1 brief (as background context only), the clinic billing matrix, and any attached image content blocks are injected into the CONSULT_EXTRACTION_PROMPT. Tools exposed: [tavily_search, emit_consult]. The model loops, may call Tavily once for a drug-recall check, then commits via emit_consult.
- **Feature 3 (Triage, Sonnet 4.6):** owner Telegram message, patient metadata (species, age, procedure, days post-visit), any owner-attached photo content blocks, and the last 5 clinic-specific corrections are injected into the TRIAGE_PROMPT. Tools exposed: [tavily_search, request_photo, request_temperature, request_appetite_timeline, request_medication_compliance, schedule_doctor_callback, emit_decision]. The model may invoke ONE clarifying tool to ask the owner; the next reply re-enters the loop with toolCallCount=1, after which the model MUST call emit_decision.

2.1.3 Dependency Diagram

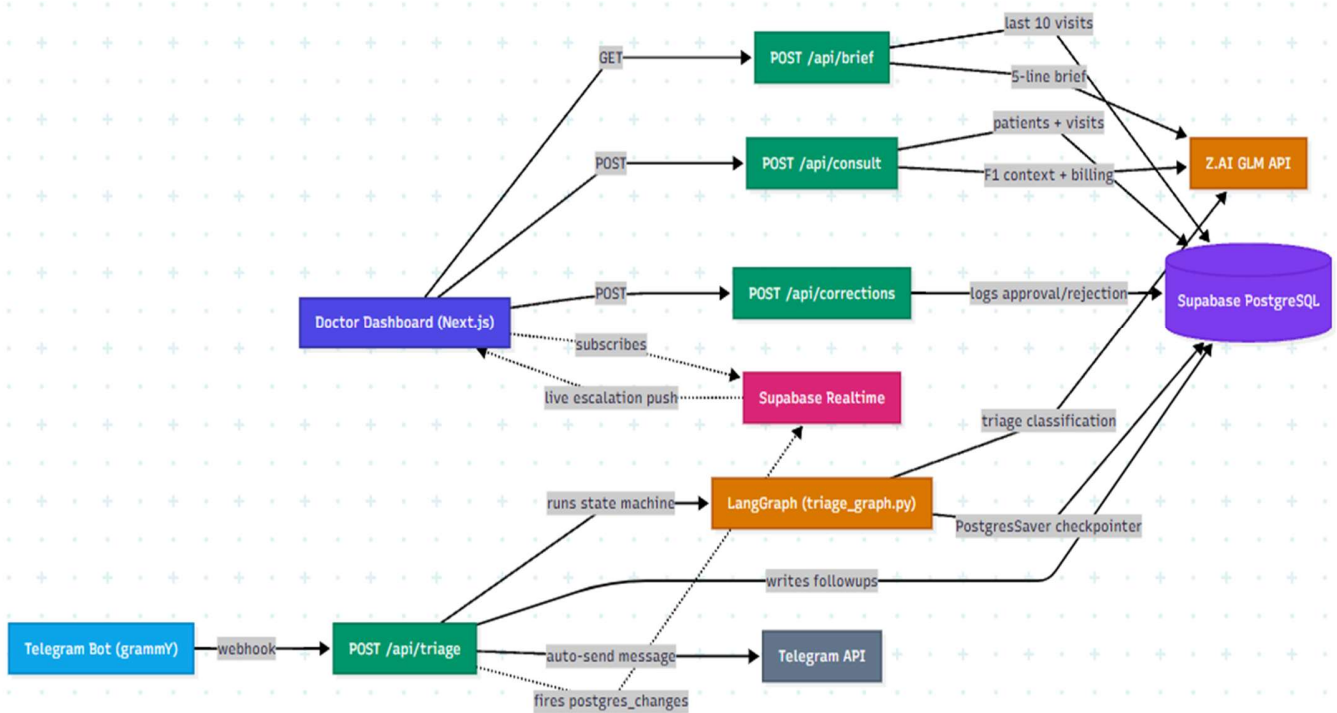
The following describes how all system components interact.

Context window management:

To prevent token overflow and control cost, the following constraints are enforced before data reaches Claude:

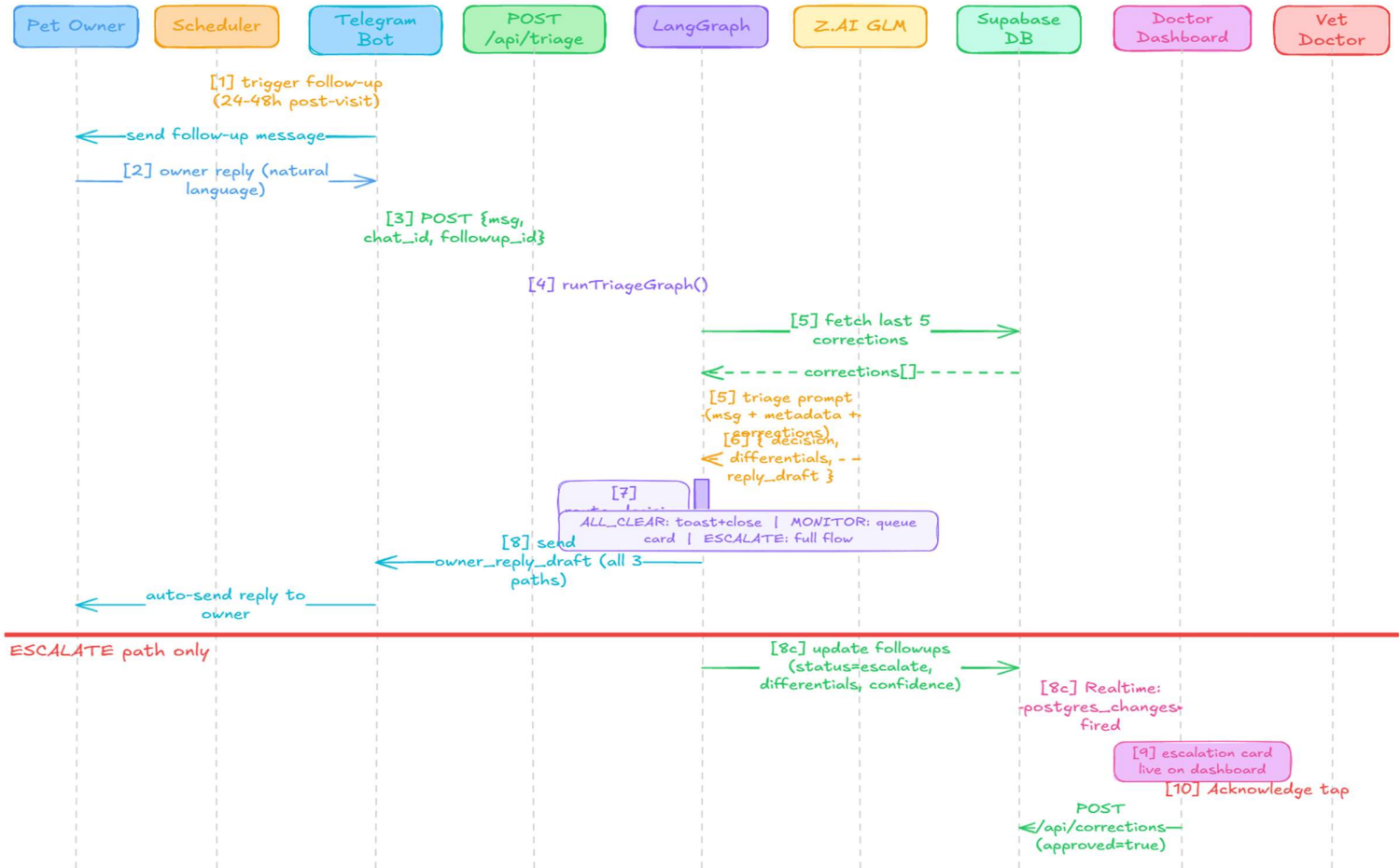
- **Feature 1 (Brief):** capped at last 10 visits; raw_notes concatenated with visit-date separators.

- **Feature 2 (Consult):** F1 brief injected as a framed background context block, explicitly instructed not to contribute findings unless referenced in the doctor's notes. Image blocks are passed by URL when Supabase Storage is configured, or as inline base64 fallback.
- **Feature 3 (Triage):** owner message (natural language, typically short), patient metadata (structured, compact), and last 5 corrections (few-shot examples, each under 100 tokens). toolCallCount strictly capped at 1 to bound the loop.
- **Tavily cache:** 7-day TTL keyed on normalised query text — prevents repeat web calls for the same drug name across follow-ups.
- **Server-side only:** all Claude, Deepgram, and Tavily calls are server-side — no token usage on the client. Timeouts set at 10 seconds with a loading state surfaced to the user.



2.1.4 Sequence Diagram

Hero Flow: Visit to Escalation Acknowledgement



2.2 Technology Stack

2.2.1 Frontend

Technology	Usage	Justification
Next.js 16 (App Router) + React 19	Doctor Dashboard, Pet Passport public page, all API routes	Single codebase for frontend and backend. API routes are server-side — Anthropic, Deepgram, and Tavily keys never reach the client. One-command Vercel deployment.

Technology	Usage	Justification
Three.js (r184)	3D animated dog mascot on landing page hero section	Cursor-tracking head, wagging tail, idle sway — used as a live product signal display on the landing page.
Supabase Realtime (client)	Live subscription on followups table in dashboard	Escalation cards appear without page refresh. Fires on postgres_changes event filtered to status=eq.escalate.
Browser MediaRecorder	Captures consult-room audio for Deepgram transcription	Native Web API, no extra dependency. Three-state UI: Record voice → Recording → Transcribing.

2.2.2 Backend

Technology	Usage	Justification
Next.js API Routes	All REST endpoints: /api/brief, /api/consult, /api/triage, /api/transcribe, /api/upload, /api/corrections, /api/patients	Co-located with frontend in one repo. No CORS configuration. Server-side execution protects API keys.
Anthropic Claude (Haiku 4.5 + Sonnet 4.6)	All AI reasoning: brief generation, multimodal consultation extraction, multi-turn triage classification with tool-use	Mandatory. Strict JSON adherence via the emit-tool pattern. Vision built in to the same model. Two-tier cost model: cheap Haiku for the brief, demo-critical Sonnet for consult and triage.
Tool Registry (lib/tools/registry.ts)	Per-feature tool exposure: server-side (tavily_search), user-facing clarifying tools (5x request_*), and emit_* structured-output tools	Each feature exposes only the tools it needs. The emit-tool pattern makes JSON parse failures impossible.
Deepgram nova-3	Voice-to-text for consult dictation	Multilingual auto-detection (English, Bahasa, Mandarin code-switching). Fast and accurate; \$200 free credit covers the hackathon.

Technology	Usage	Justification
Tavily	Server-side drug-recall and fresh-guidance lookups	1k free searches/month. Optional — when key absent, the LLM proceeds without web context. Cached in tavily_cache table for 7 days.
grammY (TypeScript)	Telegram bot — follow-up messaging, inbound owner reply handling, photo download via fetchTelegramPhotoAsImage	Runs inside a Next.js API route + standalone polling script. Multi-turn state lives in followups.conversation jsonb column.
LangGraph (Python sidecar)	Triage state machine with Supabase PostgresSaver checkpointer (DEFERRED to finals)	The TypeScript tool-use loop in lib/llm.ts handles all current flows. LangGraph is scaffolded but not on the preliminary critical path.

2.2.3 Database & Storage

Technology	Usage	Justification
Supabase (PostgreSQL)	Primary database for all application data: patients, visits, followups, corrections, tavily_cache	Managed PostgreSQL with built-in Realtime, REST API, and pub/sub. Will host LangGraph checkpointer tables alongside application data when the Python sidecar lands.
Supabase Storage	Two public buckets: consult-photos (vet-uploaded media during F2) and owner-photos (Telegram-forwarded media). Falls back to inline base64 if buckets are missing.	Public URLs let Claude vision fetch images directly. Audit trail for all uploads. The fallback ensures vision still works in any environment.
Supabase Realtime	Pub/sub on followups table for live escalation card delivery	Postgres trigger fires on row update; client subscription updates dashboard within seconds without polling.

2.2.4 Cloud & Deployment

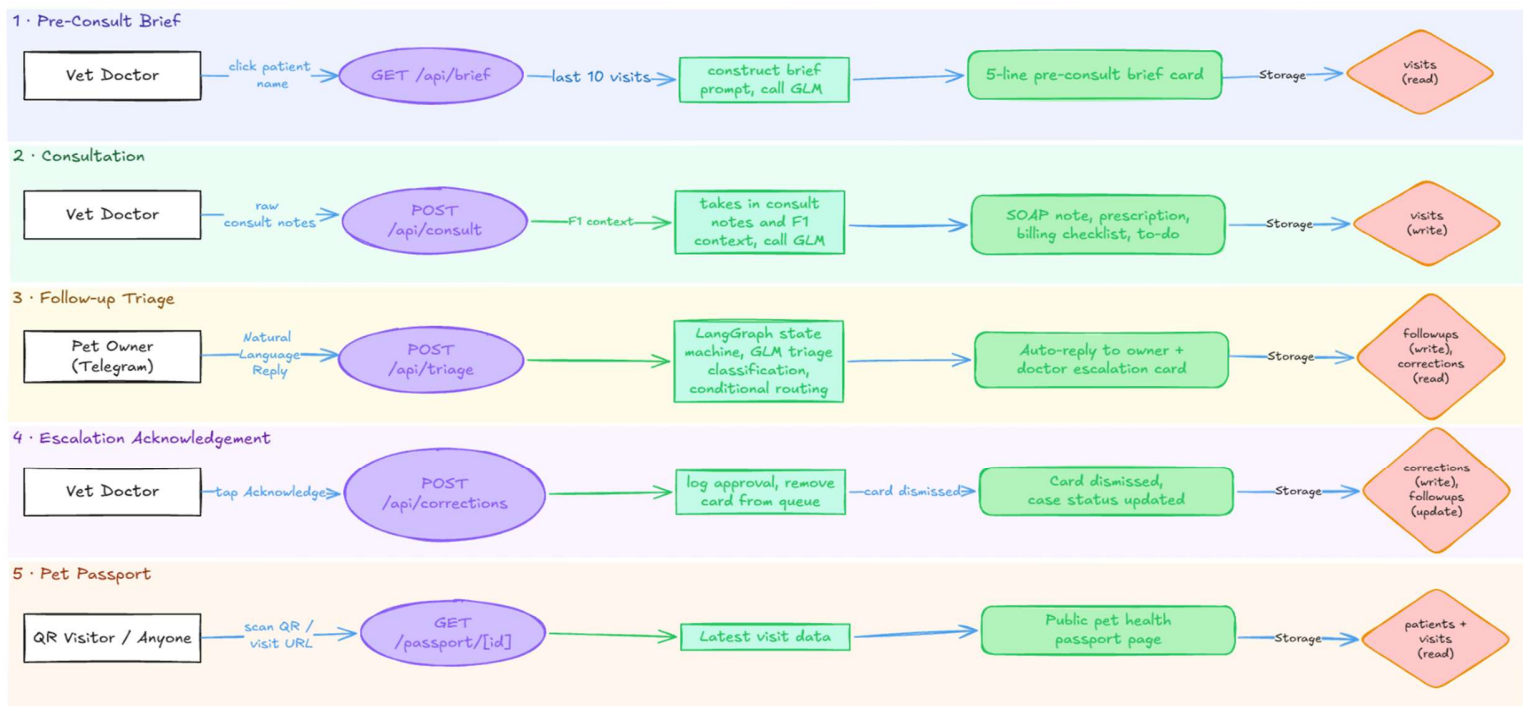
Technology	Usage	Justification
Vercel	Hosting for Next.js app and all API routes	Zero-config deployment. Serverless functions for API routes. Free tier sufficient for demo scale.
Supabase Cloud	Managed PostgreSQL + Realtime + Storage + connection pooling	No infrastructure management. Realtime pub/sub is built-in. Dashboard available for schema inspection during demo.

2.3 Key Data Flows

2.3.1 Data Flow Diagram (DFD)

The following describes data movement across all system boundaries, external entities, processes, and storage.

Data Flow Diagram (Level 1) — System Boundaries

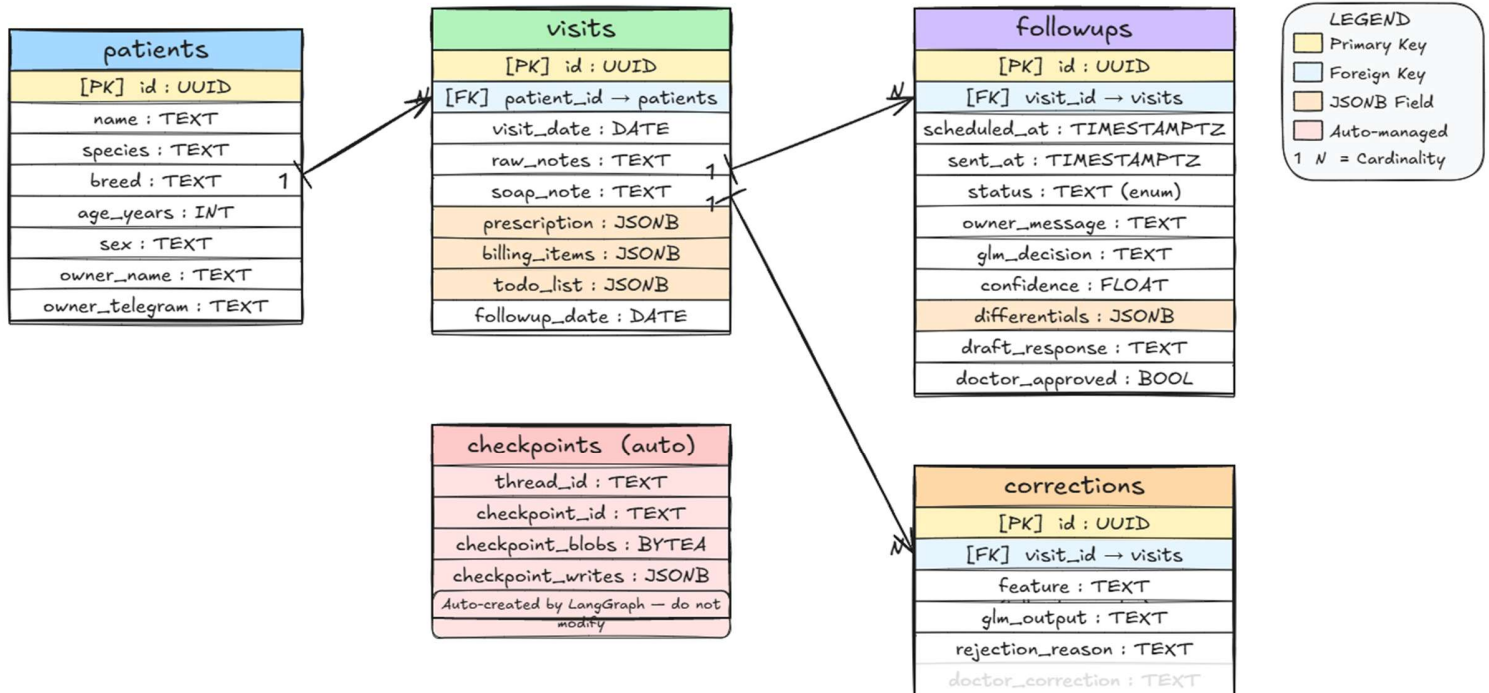


2.3.2 Normalised Database Schema (ERD — 3NF)

The schema is normalised to Third Normal Form (3NF). All non-key attributes depend solely on the primary key with no transitive dependencies. The followups table resolves the one-to-many

relationship between visits and follow-up events. The corrections table acts as a junction between visits and the feedback loop, keyed on feature type. The tavily_cache table is independent and serves as a 7-day key-value cache for web search results.

Normalised Database Schema — ERD (3NF)



3. Functional Requirements & Scope

3.1 Minimum Viable Product (MVP)

3.1.1 Core Features for Demo

#	Feature	Description
F1	Pre-Consultation Intelligence Brief	Doctor clicks a patient from today's schedule. The system fetches the patient's full visit history and calls Claude Haiku 4.5 with the [emit_brief] tool to generate a structured 5-point brief covering last visit outcome, chronic flags, owner compliance patterns, overdue pending items, and a focused probe suggestion for today's consultation.
F2	Consultation Capture & Structured Output	Doctor dictates (Deepgram nova-3) or types notes after the consult and may attach wound, lab, or X-ray photos. Claude

#	Feature	Description
		Sonnet 4.6 generates a SOAP note, a structured prescription, a billing recovery checklist (flagging items mentioned but not yet billed), and a staff to-do list — all via a single <code>emit_consult</code> tool call. The model may call <code>tavily_search</code> once if a recommended drug needs a recall check. Doctor reviews and approves each output card individually.
F3	AI Follow-Up Triage via Telegram	24–48 hours post-visit, the system auto-messages the owner. The owner's natural-language reply (with optional photo) is processed by a Claude Sonnet 4.6 tool-use loop. The model may invoke exactly ONE clarifying tool (<code>request_photo</code> , <code>request_temperature</code> , <code>request_appetite_timeline</code> , <code>request_medication_compliance</code> , <code>schedule_doctor_callback</code>) to ask the owner a targeted question, after which it MUST commit a decision via <code>emit_decision</code> : <code>ALL_CLEAR</code> , <code>MONITOR</code> , or <code>ESCALATE</code> . <code>ALL_CLEAR</code> and <code>MONITOR</code> auto-send appropriate replies; <code>ESCALATE</code> additionally creates an urgent acknowledgement card on the dashboard.
F4	Doctor Dashboard with Realtime Escalations	Three-column morning view: today's schedule (with expandable briefs), follow-up queue (colour-coded by triage level), and monthly KPI cards (time saved, billing recovered, complications caught, follow-up rate). Escalation cards appear live via Supabase Realtime without page refresh.
F5	Prompt-Based Feedback Loop	Doctor taps one-tap feedback (Correct / Wrong with reason tag) on any Claude output. Rejections are logged to the corrections table. Before each subsequent Claude call of the same feature type, the last 5 corrections are fetched and injected as clinic-specific few-shot examples, making the system progressively smarter per clinic without model retraining.
F6	Pet Health Passport	Auto-generated after every visit. Accessible via a public QR link (<code>consilium.app/passport/{uuid}</code>). Displays vaccination status, active medications, last diagnosis, and vet notes. Requires no extra doctor action. Serves as organic product discovery when any vet scans the QR.

3.2 Non-Functional Requirements (NFRs)

Quality	Requirement	Implementation
Triage Accuracy	Claude triage classification must exceed 90% accuracy against a keyword-rule baseline (~61%).	Tool-use loop with confidence-scored differentials and one optional clarifying turn. Clinic-specific corrections injected as few-shot context before each triage call.
Token Latency	Claude calls must return within 5 seconds under normal load (p95).	Server-side API routes with async calls. Per-feature model routing: Haiku for the brief, Sonnet for consult/triage. User-facing loading states (skeleton cards) displayed while awaiting Claude response. 10-second hard timeout with graceful degradation.
Realtime Responsiveness	Escalation cards must appear on dashboard within 2 seconds of triage decision being written.	Supabase Realtime subscription on followups table filtered to status=eq.escalate. Postgres trigger fires on row update.
Cost Efficiency	Average Claude token cost per feature call must remain within defined budget thresholds.	Inputs are scoped: F1 capped at 10 visits, F2 notes are single-consultation only, F3 triage context is compact. Corrections capped at last 5. toolCallCount capped at 1. Tavily results cached 7 days. No unbounded context windows.
Data Integrity	Structured Claude outputs (SOAP, prescription, billing, decision) must conform to schema before database insertion.	All structured outputs are produced via emit_* tool calls. The tool input schema is enforced by Anthropic's tool-use, so JSON parse failures are impossible.
Reliability	Follow-up messages must not be sent more than once per followup record.	followups.status field acts as idempotency guard. Scheduler checks status=pending before triggering. Telegram send only fires once per followup_id.

Quality	Requirement	Implementation
Maintainability	Claude prompts, API routes, and tool registry must be independently modifiable.	Prompts centralised in lib/prompts.ts. Tool registry centralised in lib/tools/registry.ts. API routes are thin — they call lib functions, not inline logic.
Observability	Every Claude interaction must be reproducible from logs.	All Claude requests log: feature, model, prompt hash, tool calls, tool results, final emit payload, and source (live vs fixture). Fixtures in lib/glm-fixtures.ts allow deterministic demos.

3.3 Out of Scope / Future Enhancements

The following represent the product's grand vision but are explicitly out of scope for UMHackathon 2026:

- Twilio voice calls — Telegram is sufficient and widely trusted in SEA.
- Inventory management — generic and crowded space, not a decision intelligence problem.
- Owner-facing mobile app — Telegram is the app; no new downloads required.
- Full PMS integration — CSV import used for demo; full integration is a post-hackathon roadmap item.
- Payment processing — out of scope for this system's decision layer.
- User authentication — single hardcoded clinic for demo purposes.
- Multi-clinic enterprise features — post-hackathon growth story.
- LangGraph PostgresSaver checkpointing — scaffolded but deferred to finals; the TypeScript tool-use loop covers all preliminary flows.

4. Monitoring, Evaluation, Assumptions & Dependencies

4.1 Technical Evaluation

4.1.1 Grayscale Rollout & A/B Testing

Post-hackathon deployment strategy: new Claude prompt versions (e.g. an updated triage prompt) are released to 10% of clinic sessions initially. Triage accuracy, doctor correction rate, and escalation false-positive rate are monitored over a 48-hour window. If no regression is detected, the prompt version is promoted to 100%. A/B testing is conducted by comparing doctor correction rates between the incumbent and challenger prompt versions; the variant with the lower correction rate is promoted. The two-tier model routing (Haiku 4.5 vs Sonnet 4.6) is also configurable per environment via `ANTHROPIC_MODEL_BRIEF`, `ANTHROPIC_MODEL_CONSULT`, and `ANTHROPIC_MODEL_TRIAGE` — enabling cheap-tier experiments without code changes.

4.1.2 Emergency Rollback (ER) & Golden Release

If the triage false-escalation rate exceeds 20% in a rolling 24-hour window, an automated pipeline reverts the prompt template (`lib/prompts.ts`), the tool registry (`lib/tools/registry.ts`), and the model routing config to the last validated Golden Release. The corrections table is preserved across rollbacks — clinic-specific learning is not lost. For the hackathon demo, a seeded Supabase snapshot is treated as the Golden baseline and can be restored in under 2 minutes via Vercel redeploy. The mock-mode fixtures in `lib/glm-fixtures.ts` also serve as a deterministic fallback if the live Anthropic API is unavailable during the demo.

4.1.3 Priority Matrix

Priority	Trigger Condition	Action
P1 — Critical	Anthropic API returns errors on >5% of calls in a 10-minute window.	Immediately surface AI Unavailable state on dashboard. Disable auto-triage; Telegram follow-ups queued but not sent. Fall back to <code>lib/glm-fixtures.ts</code> mock mode for the demo. Alert on-call developer. Rollback to Golden Release if unresolved within 30 minutes.
P1 — Critical	Escalation card fails to appear on dashboard within 10 seconds of an	Investigate Supabase Realtime subscription health. Force-refresh dashboard subscription. Log incident.

Priority	Trigger Condition	Action
	emit_decision write with status=escalate.	
P2 — High	Doctor correction rate on triage exceeds 30% in a single day.	Flag for prompt review. Pause few-shot injection temporarily. Escalate to manual triage review for affected cases.
P2 — High	Telegram message fails to deliver to owner after 3 retry attempts.	Mark followup.status = failed. Surface a manual outreach task in the doctor's to-do queue. Log failure with owner_telegram and followup_id.
P2 — High	Deepgram transcription failure rate exceeds 5% in a rolling hour.	Surface raw audio playback to the doctor. Fall back to typed input. Log audio mime types and lengths for Deepgram support ticket.
P3 — Medium	emit_consult tool input fails Zod schema validation post-call (rare — should not happen with tool-use).	Return error state to frontend. Doctor sees Generation failed — please re-submit. Raw notes preserved in database. Log malformed Claude response for prompt debugging.

4.2 Monitoring

The following observability strategy ensures system health is continuously tracked:

Metric	Source	Alert Threshold	Response
Anthropic API error rate	Server-side API route error logs (Vercel)	>5% errors in 10 min	P1 escalation — graceful degradation state, fallback to fixtures
Triage decision latency	lib/llm.ts callGLM timing logs	>8 seconds p95	Investigate Claude throughput; consider prompt length reduction or model swap
Tool-use loop iteration count	callGLM toolCallCount metric	>2 per triage case (should always be ≤ 1)	Investigate prompt; clarifying tool may be over-triggering

Metric	Source	Alert Threshold	Response
Doctor correction rate per feature	corrections table (Supabase)	>30% corrections/day for any feature	P2 — prompt review triggered
Realtime event delivery lag	Supabase Realtime dashboard	>3 seconds lag	Investigate DB replication lag; restart subscription
Telegram delivery failures	grammY error handler	>3 consecutive failures per owner	Mark followup failed; surface manual task to doctor
Deepgram transcription error rate	/api/transcribe response logs	>5% errors in rolling hour	Fallback to typed input; log audio specs
Tavily cache hit rate	tavily_cache row inserts vs Tavily API calls	<50% hit rate sustained	Review query normalisation logic; cache may be over-fragmenting
Follow-up response rate	followups table — replied vs sent ratio	<50% response rate in 7-day window	Review message timing and copy; flag for product iteration

4.3 Assumptions

The following operational and environmental conditions are assumed to be valid during development and deployment:

- Vet doctors have a stable internet connection and access to a desktop or tablet browser to use the dashboard during clinic hours.
- Vet doctors' browsers support the MediaRecorder API for voice dictation. Typed-note fallback is provided when not.
- Pet owners have a Telegram account linked to the phone number recorded in patients.owner_telegram.
- The clinic operates on a single-timezone schedule (no multi-timezone follow-up scheduling required).
- Consultation notes and Telegram replies are provided in English, Bahasa Malaysia, or Mandarin — Claude (and Deepgram nova-3 for audio) handle code-switching common in SEA clinical contexts.

- The clinic's billing matrix is seeded as a static JSON config file at deployment time (not a dynamic pricing engine).
- For the hackathon demo, a single hardcoded clinic is assumed with 150 synthetic patients seeded via seed.sql.
- The Anthropic API key, Deepgram API key, and (optional) Tavily API key are configured as Vercel environment variables before deployment.
- Supabase Storage buckets consult-photos and owner-photos are created and public; if not, the system falls back to inline base64 image passing.

4.4 External Dependencies

Tool / Service	Purpose	Risk Level	Mitigation Strategy
Anthropic Claude API (Haiku 4.5 + Sonnet 4.6)	Core reasoning engine for all three Claude features: brief generation, multimodal consultation extraction, and triage classification with tool-use. Non-removable — the system's value proposition depends entirely on its clinical reasoning capability.	HIGH — entire core product fails if rate-limited or unavailable.	Implement graceful degradation: surface AI Unavailable banner, disable auto-triage, queue follow-ups. Fall back to lib/glm-fixtures.ts for deterministic demo paths. Cache last successful brief per patient for re-display. Alert developer immediately.
Deepgram nova-3	Voice-to-text for consult dictation.	LOW — typed notes are a fully functional fallback.	On Deepgram error, surface raw audio playback and the typed-note textarea. The consult flow continues without transcription.
Tavily Web Search	Server-side drug-recall and fresh-guidance lookups during F2 and F3.	LOW — optional. When the Tavily key is absent or the API errors, Claude proceeds without web context.	7-day cache (tavily_cache) absorbs repeat queries. Missing-key path is unit-tested. The model's prompt explicitly tolerates absent web context.
Supabase (PostgreSQL + Realtime + Storage)	Primary database, live event bus for escalation cards, and image hosting for consult-photos and owner-photos.	MEDIUM — if Realtime is disrupted, escalation cards	Implement polling fallback (30-second interval) on dashboard if Realtime subscription drops. All data is persisted regardless of

Tool / Service	Purpose	Risk Level	Mitigation Strategy
		require manual refresh. If Storage buckets are missing, image passing falls back to inline base64.	Realtime health. lib/storage.ts already implements the base64 fallback.
Telegram Bot API	Owner-facing follow-up interface. All post-discharge communication flows through Telegram, including owner-sent photos.	MEDIUM — if Telegram API is unavailable, follow-ups cannot be sent.	Queue unsent follow-ups in the followups table (status=pending). Retry on the next scheduler cycle. Surface pending queue to doctor as manual outreach list if retries fail.
Vercel	Hosting for the Next.js app and all serverless API routes.	LOW — Vercel has 99.99% uptime SLA.	Demo snapshot committed to Git. Can redeploy in under 2 minutes from any machine.
LangGraph (Python sidecar)	Triage state machine with PostgresSaver for conversation continuity (DEFERRED to finals).	LOW — not on the preliminary critical path.	TypeScript tool-use loop in lib/llm.ts handles all preliminary flows. LangGraph remains scaffolded for finals-round multi-turn checkpointing.

5. Project Management & Team Contributions

The preliminary round spans approximately 10 days, structured as a two-week sprint with a demo at the end of Day 10.

5.1 Project Timeline

Day(s)	Phase	Deliverables
Day 1–2	Planning & Architecture	Finalise SAD, lock DB schema, set up Supabase project (with consult-photos and owner-photos buckets), seed 150 synthetic patients, configure Vercel deployment pipeline, verify Anthropic / Deepgram / Tavily keys.

Day(s)	Phase	Deliverables
Day 2–3	Database & API Foundations	Implement patients and visits CRUD routes, create Claude wrapper (lib/llm.ts) and tool registry (lib/tools/registry.ts), establish Supabase client and Realtime subscription baseline, wire Tavily executor with tavily_cache.
Day 3–4	F1 — Pre-Consult Brief	Implement GET /api/brief, BRIEF_PROMPT with [emit_brief] tool, PatientCard component with expandable brief, integrate into Dashboard schedule column.
Day 4–5	F2 — Consultation Capture (multimodal)	Implement POST /api/consult, POST /api/transcribe (Deepgram nova-3), POST /api/upload (Supabase Storage), CONSULT_EXTRACTION_PROMPT with [tavily_search, emit_consult], ConsultCapture component with voice + photo + text input and four output cards.
Day 5–7	F3 — Triage & Telegram (multimodal)	Implement TRIAGE_PROMPT with the full tool registry (5 clarifying tools + tavily_search + emit_decision), POST /api/triage, lib/telegram-handler with photo download via fetchTelegramPhotoAsImage, grammY bot follow-up logic, Realtime subscription, EscalationCard modal.
Day 7–8	F4 Dashboard + F5 Feedback Loop	Assemble three-column Dashboard, FollowUpQueue with colour coding, MetricsRow KPI cards, /api/corrections route, one-tap feedback UI, few-shot injection on subsequent Claude calls.
Day 8–9	F6 Pet Passport + Landing Page	Implement /passport/[id] public page, QR generation, HeroDog Three.js (r184) component with cursor tracking, landing page layout.
Day 9–10	Integration, Polish & Demo Prep	End-to-end flow testing, seed realistic demo scenarios (escalation, monitor, all_clear, ambiguous-with-photo), polish UI, prepare demo script and walkthrough, lock fixtures in lib/glm-fixtures.ts as a deterministic fallback.

5.2 Team Members & Roles

Member	Role	Primary Responsibilities
Brandon	AI Engineer	Anthropic Claude integration (lib/llm.ts), tool registry (lib/tools/*), Tavily wiring with 7-day cache, all prompt chains (lib/prompts.ts), TypeScript tool-use loop, LangGraph triage graph (deferred to finals), feedback loop few-shot injection, fixtures (lib/glm-fixtures.ts).
Zi Qian	Software Engineer	Next.js 16 setup, all API routes (/api/brief, /api/consult, /api/triage, /api/transcribe, /api/upload, /api/corrections, /api/patients), Supabase schema and seed, Telegram bot (grammY) including photo download, Supabase Realtime wiring, Vercel deploy.
Yu Han	Data Analyst	150 synthetic patients (supabase/seed.sql) across 10 diagnoses and 3 recovery patterns, 50 Telegram reply scenarios (20 normal / 15 ambiguous / 15 red-flag), validation accuracy table (Claude vs keyword baseline), dashboard metric simulations.
Shawn	Frontend + PM	All dashboard screens, escalation modal (the hero screen), patient card with brief expansion, ConsultCapture component (voice + photo + text), HeroDog Three.js (r184) cursor-tracking mascot, MetricsRow KPI cards, pitch deck, demo script rehearsal.
Harrison	Domain + QA	Billing matrix (10 diagnoses × billable items mapping), realistic SOAP note scenarios for synthetic data, medical QA on all Claude outputs (SOAP, Rx, triage, billing flags), pet passport content design, end-to-end QA across all 50 Telegram scenarios.

5.3 Recommendations

The following architectural recommendations are proposed for scalability, performance, and reliability beyond the hackathon:

- **Prompt Caching** — cache the last successful F1 brief per patient_id for 24 hours. Most patients are seen once per day; regenerating the same brief on repeated dashboard loads wastes tokens. Invalidate on new visit insert.
- **Background Job Queue** — replace the ad-hoc follow-up scheduler with a proper job queue (e.g. pg_cron on Supabase or a lightweight BullMQ instance). This ensures follow-ups are sent reliably even if the server restarts between scheduling and execution.
- **Streaming Claude Responses for F2** — the consultation extraction call is the longest Claude interaction. Streaming the emit_consult tool input and progressively revealing output cards (SOAP first, then billing, then todos) would significantly improve perceived performance.
- **Corrections Table Indexing** — add a composite index on corrections(feature, created_at DESC) to ensure the few-shot fetch before each Claude call remains fast as the corrections table grows.
- **Multi-Clinic Isolation** — when scaling beyond one clinic, add a clinic_id foreign key to patients, visits, followups, and corrections. Enforce row-level security (RLS) in Supabase so each clinic's data is isolated by default.
- **Claude Model Fallback** — the per-feature model routing (ANTHROPIC_MODEL_BRIEF / _CONSULT / _TRIAGE) already enables hot-swap. Configure a secondary tier (e.g. Haiku 4.5 as fallback for Sonnet 4.6) to gracefully degrade if the primary tier is rate-limited, with a lower-confidence indicator surfaced to the doctor.
- **LangGraph Activation** — promote the deferred Python sidecar to production for finals. The PostgresSaver checkpointer enables true multi-day conversation continuity (e.g. day-1 owner reply → clarifying photo request → day-2 owner reply with photo → final decision) where the current TypeScript loop is single-turn-windowed.